

✓ Assignment - 1 ✓

Q.1: Let H be a finite non-empty subset of a group G . If H is closed under multiplication then H is a subgroup of G .

Sol: Let $H (\neq \emptyset) \subseteq G$ be finite. Also, H is closed under multiplication (*) i.e. for $h_1, h_2 \in H$ then $h_1 * h_2 \in H$.

Existence of Identity: Let $a \in H$. Since, H is closed under multiplication.

Then, $a, a^2, \dots, a^n \in H$.

As H is finite, $\{a^n \mid n \in \mathbb{Z}_+\} \subseteq H$.

By using Pigeonhole principle, for some $m, n \in \mathbb{Z}_+$ ^{m and n} where $m > n$ such that

$$a^m = a^n$$

Also, $m - n > 0$, post multiplying a^{-n} which exist in G , we get

$$a \cdot a = a^{m-n} = a^n \cdot a^{-n} = e$$

$$a^{m-n} = e \in H \quad (\because m-n > 0)$$

Thus, identity is contained in H .

Existence of inverse: $\because a^{m-n} = e$

$$= a^{(m-n+1)-1} = e$$

$$= a \cdot a^{m-n+1} = e \quad (\text{where, } m-n+1 > 1)$$

$$= a^{-1} = a^{m-n-1} \in H$$

So every element of H has an inverse.

Thus, $H \leq G$.

cyclic group is cyclic.
 Let $G = \langle g \rangle$ be a cyclic group
 and let $H \leq G$.

$H = \{e\}$ then H is cyclic trivially.

Assume $H \neq \{e\}$ (non-trivial).
 Every element of G is of the form g^k for
 some $k \in \mathbb{Z}$.

Consider the set
 $S = \{k \in \mathbb{N} : g^k \in H\}$
 $H \neq \{e\}$, S is non-empty.
 By the Well-Ordering principle,

$\exists$ a positive integer $m \in S$, which is least such
 that $g^m \in H$.

Take any $h \in H$. Then, $h = g^m$ for
 some $m \in \mathbb{Z}$.

when m is divided by n ,
 $m = nq + r$, with $q \in \mathbb{Z}$,
 $\Rightarrow g^m = g^{nq+r} = (g^n)^q \cdot g^r$
 $0 \leq r < n$

But n was the least
 so $r=0$ necessarily.

$\therefore g^m = g^{nq} \in H$ ($\because g^n \in H$)

Thus, m is a multiple of n .

and $g^m \in \langle g^n \rangle$. Thus, $h \in \langle g^n \rangle$.

Hence, $H = \langle g^n \rangle$, so H is cyclic.

Q.3: Let G be a finite cyclic group of order n .
 Show that for every divisor d of n , G has
 a unique subgroup of order d .

Sol: Let $G = \langle a \rangle$ such that $|G| = n$
 then,

$\exists H \leq G$, with $|H| = d$
 i.e. $d | n \Rightarrow n = dk$ for some $k \in \mathbb{Z}$
 $\therefore H = \langle a^k \rangle$ with $O(a^k) = d$.

$$\text{as, } O(a^k) = \frac{n}{\gcd(n, k)} = \frac{n}{k}$$

Now, let T be a subgroup of G , $|T| = d$.

Suppose $T = \langle a^r \rangle$

$$\therefore |T| = O(a^r) = d$$

$$\Rightarrow a^{rd} = e$$

$$\therefore n | rd$$

$$\therefore rd = np \text{ for some } p \in \mathbb{Z}$$

$$\Rightarrow rd = kp \text{ (using eq(i))}$$

$$\Rightarrow r = kp \Rightarrow a^r = a^{kp} = (a^k)^p$$

$$\text{i.e. } a^r = (a^k)^p \in H$$

$$\Rightarrow \boxed{T=H} \text{ i.e. } G \text{ has a unique subgroup}$$

Q.4: Compute:

$$(a) \text{ Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$$

Then, $\text{Hom}(G, H) \cong \{x \in H \mid mx = 0 \text{ mod } n\}$
 $\therefore$ any homomorphism $f: \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$
 is determined by the image of 1, and that image must satisfy

$$f(m) = mf(1) = 0 \text{ in } H.$$

So we're looking for elements $x \in \mathbb{Z}/n\mathbb{Z}$ such that $mx \equiv 0 \text{ mod } n$.

$$(a) \text{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z})$$

$$\therefore x \in \mathbb{Z}/6\mathbb{Z} \text{ such that } 4x \equiv 0 \text{ mod } 6$$

$$x=0, \quad 4 \cdot 0 = 0 \text{ mod } 6 \quad \checkmark$$

$$x=1, \quad 4 \cdot 1 = 4 \text{ mod } 6 \quad \times$$

$$x=2, \quad 4 \cdot 2 = 8 \text{ mod } 6 \quad \times$$

$$x=3, \quad 4 \cdot 3 = 12 \text{ mod } 6 \quad \checkmark$$

$$x=4, \quad 4 \cdot 4 = 16 \text{ mod } 6 \quad \times$$

$$x=5, \quad 4 \cdot 5 = 20 \text{ mod } 6 \quad \times$$

$$x=6, \quad 4 \cdot 6 = 24 \text{ mod } 6 \quad \checkmark$$

So the valid values are $x=0, 3$

There are two homomorphism.

$$\therefore \text{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/6\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

$$(b) \text{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/7\mathbb{Z}) \cong \{0\} \text{ (trivial group)}$$

$$\text{Let } x \in \mathbb{Z}/7\mathbb{Z} \text{ such that } 4x \equiv 0 \text{ mod } 7$$

$$x=0 \quad \checkmark \quad x=3 \quad \times \quad x=6 \quad \times$$

$$x=1 \quad \times \quad x=4 \quad \times$$

$$x=2 \quad \times \quad x=5 \quad \times$$

So the valid values are $x=0$.
 There is only one single homomorphism.

$$(c) \text{Let } \text{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z})$$

$$\text{Let } x \in \mathbb{Z}$$

$$x=0, \quad \checkmark \quad f(x) = f(4 \cdot x) = 4f(x) = 0$$

$$\Rightarrow f(x) \in \mathbb{Z} \text{ such that } 4f(x) = 0$$

$\mathbb{Z}$ has no nonzero torsion elements, solution only is $f(x) = 0 \quad \forall x \in \mathbb{Z}$.

$$\therefore \text{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}) \cong \{0\}$$

Only one homomorphism.

$$\text{Similarly } \text{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\text{gcd}(m, n)$$

Q.5: Show that group of order 4 is abelian.

sol: $|G| = 4$. Let's denote.

$$G = \{e, a, b, c\} \text{ where } e \text{ is the identity.}$$

By Lagrange's Theorem

the order of any element divides the order of the group. So possible orders of non-identity elements are divisors of 4 i.e. 2 or 4.

Case I: $|a| = 4$ where $a \in G$. (Some element has order 4)
 Then, $G = \{e, a, a^2, a^3\}$. This is a cyclic group of order 4. Cyclic groups are abelian.

Case II: All non-identity elements have order 2.

$$\text{Suppose, } |a| = |b| = |c| = 2 \text{ Then, } a^2 = b^2 = c^2 = e.$$

Consider, $ab = c$ ($\because ab \neq a, ab \neq b$).

$$\text{Similarly } ba = ? \quad (\because (ab)^2 = e \Rightarrow abab = e \Rightarrow c \cdot c = e)$$

$$\therefore ba = c$$

($\because a, b, c$ are non-identity elements)

$$\Rightarrow \boxed{ab = ba}$$

Similarly check other pairs.

$\therefore$ all products commutes because of the small no. elements.

$\therefore$ if element of order 4 $\Rightarrow$ cyclic $\Rightarrow$ abelian

(if ~~clear~~ all non-identity elements are of order 2 $\rightarrow$ multiplication table shows commutative $\Rightarrow$ abelian.)

$\therefore$ Any group of order 4 is abelian.

Assignment-2

1. Show that every continuous homomorphism from $\mathbb{R}$ to itself is of the form $x \mapsto \alpha x$ for some fixed $\alpha \in \mathbb{R}$.

Sol: (I) Let $f: (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$ be a group homomorphism which is continuous which has the property, such that $f(x+y) = f(x) + f(y)$.

$\forall n \in \mathbb{Z}^{+}$, $f(n) = nf(1)$ and $f(0) = 0$ for negatives, let $m = -n \in \mathbb{Z}^{-}$
 $f(m) = mf(1) \Rightarrow f(-n) = -nf(1)$

Clearly $\forall x \in \mathbb{Z}$, $f(x) = \alpha x$ for some fixed $\alpha \in \mathbb{R}$.

(II) Suppose, $\frac{p}{q} \in \mathbb{Q}$, then

$$f\left(\frac{p}{q}\right) = f\left(\underbrace{\frac{1}{q} + \frac{1}{q} + \dots + \frac{1}{q}}_{p \text{ times}}\right) = f\left(\frac{1}{q}\right) + f\left(\frac{1}{q}\right) + \dots + f\left(\frac{1}{q}\right) \\ = pf\left(\frac{1}{q}\right) = \frac{p}{q} \cdot 2f\left(\frac{1}{2}\right)$$

$$= \frac{p}{q} \cdot f\left(\underbrace{\frac{1}{2} + \frac{1}{2} + \dots + \frac{1}{2}}_{q \text{ times}}\right) = \frac{p}{q} f(1)$$

Clearly $\forall x \in \mathbb{Q}$, $f(x) = \alpha x$ for some fixed $\alpha \in \mathbb{R}$.

Let $\forall x \in \mathbb{Q}$

(III) Suppose $c \in \mathbb{Q}^c$ then

$\exists$ sequence of rationals such that

$$(x_n) \rightarrow c \text{ \& } f(x_n) = \alpha x_n \text{ for some fixed } \alpha \in \mathbb{R} \\ \text{i.e. } \lim_n x_n = c$$

$\because f$ is continuous, $\Rightarrow \lim_n f(x_n) = f(c)$

$$= \lim_n \alpha x_n = f(c) \Rightarrow \boxed{\alpha c = f(c)}$$

Clearly $\forall x \in \mathbb{Q}^c$, $f(x) = \alpha x$ for some fixed $\alpha \in \mathbb{R}$.

Q.2: Show that any finite subgroup of $\mathbb{C}^*$ is cyclic.

Sol: Let

$$S = \{x \in \mathbb{C}^* \mid x^n = 1\}$$

$$= S = \left\{ e^{\frac{2\pi i k}{n}} \mid 0 \leq k \leq n-1 \right\}$$

$$= S = \{ \alpha^k \mid 0 \leq k \leq n-1 \} \text{ where } \alpha = e^{\frac{2\pi i}{n}}$$

Now, clearly any number in the set S is of the form, $\langle \alpha^k \rangle$ which is cyclic

Now let $H \subseteq \mathbb{C}^*$ such that $|H| = n$ (finite)
take any $a \in H$ then, $a^n = 1 \Rightarrow a \in S$.
So, $H \subseteq S$.

2. Let G be an abelian group which has elements of order m, n . $\text{lcm}(m, n)$.
 Show that G has an element of order $\text{lcm}(m, n)$.

Ans: Let $a, b \in G$ such that $O(a) = m$, $O(b) = n$
 where $m = p_1^{e_1} p_2^{e_2} \dots p_r^{e_r}$; $n = p_1^{f_1} p_2^{f_2} \dots p_r^{f_r}$

Example $O(a) = 60 = 2^2 \cdot 3 \cdot 5 \cdot 7$; $O(b) = 630 = 2 \cdot 3^2 \cdot 5 \cdot 7$
 $\therefore \text{LCM}(m, n) = p_1^{\max\{e_1, f_1\}} p_2^{\max\{e_2, f_2\}} \dots p_r^{\max\{e_r, f_r\}}$

$= k_1 \cdot k_2$, where k_1 is the product of all those with $e_i \geq f_i$ and k_2 is the product of all those with $e_i < f_i$

Eg: $\text{LCM}(60, 630) = \underbrace{2^2 \cdot 5 \cdot 7}_{k_1} \cdot \underbrace{3^2}_{k_2}$

so, $k_1 = \prod_{e_i \geq f_i} p_i^{e_i}$; $k_2 = \prod_{e_i < f_i} p_i^{f_i}$

$\text{lcm}(k_1, k_2) = k_1 \cdot k_2$ & $\text{gcd}(k_1, k_2) = 1$

Clearly, $k_1 \mid m$ and $k_2 \mid n$

$\frac{m}{k_1}, \frac{n}{k_2} \in \mathbb{Z}$

if $O(a^{m/k_1}) = k_1$ & $O(b^{n/k_2}) = k_2$

$O(a^{m/k_1} \cdot b^{n/k_2}) = k_1 \cdot k_2 = \text{lcm}(m, n)$
 ($\because G$ is abelian).

In the group $G = \mathbb{C}^*$, find the cosets of the subgroup $H = \{z \in G \mid |z| = 1\}$ and describe them geometrically.

Ans: We know that a left coset of H in G is of the form:

$zH = \{zh : h \in H\}$ for some fixed $z \in \mathbb{C}^*$.

$\because h \in H$ has modulus 1.

Let's write $z = re^{i\phi}$ with $r = |z| > 0$. Then

$$zH = \{re^{i\phi} \cdot e^{i\theta} = re^{i(\phi+\theta)} : \theta, \phi \in \mathbb{R}\}$$

This is the set of all complex numbers with modulus r , i.e. a circle of radius r centred at the origin and with argument $\theta + \phi$

$\therefore$ subgroup H is itself a unit circle and. Also, two elements $z_1, z_2 \in \mathbb{C}^*$ lie in the same coset iff $|z_1| = |z_2|$

Geometrically, cosets are concentric circles about the origin, one circle for each positive radius.

Q.5: Let H be a subgroup of G such that $x^2 \in H \forall x \in G$. Show that H is a normal subgroup and G/H is abelian.

Solⁿ: Let $H \leq G$

Take any $h \in H$ and $g \in G$.

(I) To show that H is a normal subgroup,

i.e. $ghg^{-1} \in H$

Consider, $ghg^{-1} = ghgh^{-1}g^{-1}g^{-1}$
 $= (gh)(gh)h^{-1}(g^{-1})(g^{-1})$
 $= (gh)^2 h^{-1} (g^{-1})^2$

$\because (gh)^2 \in H, h^{-1} \in H, (g^{-1})^2 \in H$. ($\because gh \in H$).

$\therefore ghg^{-1} \in H$.

So, H is a normal group.

i.e. $(xH)(yH) = (yH)(xH)$

We want to show that all commutators $[x, y] = xyx^{-1}y^{-1} \in H$, which would imply G/H is abelian.

Let $x, y \in H$.

Consider $xyx^{-1}y^{-1} = (xy)(xy^{-1}x^{-1}y^{-1})$
 $= (xy)^2 y^{-1} (x^{-1})^2 y^{-1}$

Consider,

$$xyx^{-1}y^{-1} = (xy)(xyy^{-1}x^{-1}yy^{-1}x^{-1}y^{-1})y^{-1}$$

$$= (xy)^2 (y^{-1}x^{-1}y)^2 y^{-2} \in H \checkmark$$

Clearly $[x, y] \in H$.

$\therefore \frac{G}{H}$ is abelian $\left(\begin{array}{l} \because ab^{-1} \in H \\ \Leftrightarrow aH = bH. \end{array} \right.$
 where $a = xy, b = yx$

$\therefore (xH)(yH) = (yH)(xH)$

6. Let G be a finite group. Suppose

$\forall x \in G$, there exists $y \in G$ such that $y^2 = x$, then G has odd order, and conversely.

Solⁿ $\Leftarrow$ Suppose, $|G| = 2m+1$

if $x \in G$ then $x^{2m+1} = e$
 $x^{2m} = x^{-1}$
 $x = (x^{-m})^2 = y^2$, where $y = x^m \in G$

$\Rightarrow$ Let $|G| = 2m$ for the sake of contradiction
 $\forall x \in G, x^{2m} = 1$

$\Rightarrow$ Let $|G| = \text{even}$ then $\exists a \in G$ such that $O(a) = 2$.

Consider, $f: G \rightarrow G$ with $f(x) = x^2$ and

$f(1) = 1, f(a) = a^2 = 1$

Clearly, f is not 1-1.

$\Rightarrow f$ is not surjective ($\because G$ is finite)

i.e. $\exists$ some $y \in G$ such that $y \neq f(x)$

$\Rightarrow x^2 \neq y \forall x \in G$. which is a contradiction.

$\therefore |G|$ is odd.

Assignment-3

Q.1: Prove that the quotient group $\mathbb{R}/\mathbb{Z}$ is isomorphic to the circle (with respect to multiplication).

Sol: Define $f: \mathbb{R} \rightarrow S^1$
 $x \mapsto e^{2\pi i x}$

Now, $\text{Ker } f = \{x \in \mathbb{R} \mid f(x) = 1\}$

$= \{x \in \mathbb{R} \mid e^{2\pi i x} = 1\}$

$= \{x \in \mathbb{R} \mid 2\pi i x = 2\pi i n \quad \forall n \in \mathbb{Z}\}$

$= \{x \in \mathbb{R} \mid x = n \in \mathbb{Z}\}$

$= \mathbb{Z}$

By first isomorphism theorem,

$\mathbb{R}/\text{ker } f = \mathbb{R}/\mathbb{Z} \cong S^1$

Q.2: Let H and K be subgroups of a group G of finite index. Show that $H \cap K$ also has finite index.

Solution: $[G : (H \cap K)] \leq [G : H] \cdot [G : K]$
 So, $[G : (H \cap K)]$ is finite whenever $[G : H]$ and $[G : K]$ is finite. (To prove).

$$\begin{aligned} \text{LHS} &= a(H \cap K) \\ &= \{at \mid t \in H \cap K\} \\ &= \{at \mid t \in H \text{ and } t \in K\} \\ &= \{at \mid t \in H\} \text{ and } \{at \mid t \in K\} \\ &= aH \cap aK = \text{RHS} \end{aligned}$$

~~$\therefore [aH] = [G : H]$ is finite &~~

$\therefore aH$ is finite and aK is finite

$\Rightarrow aH \cap aK = a(H \cap K)$ is finite

$$\text{Thus, } [G : H \cap K] \leq [G : H] \cdot [G : K]$$

Q.3: Let G be a finite group and H and K are subgroups of G .
 Prove that:

$$|HaK| = \frac{|H||K|}{|H \cap aKa^{-1}|} \quad \forall a \in G$$

By using the result,

If $K \leq G$ then for $g \in G$

$$(i) \quad gKg^{-1} \leq G$$

$$(ii) \quad |gKg^{-1}| = |K| \quad \left(\text{Prove } f: K \rightarrow gKg^{-1} \right. \\ \left. k \mapsto gkg^{-1} \right)$$

$$\text{Assume, } |HaK| = |HaKa^{-1}|$$

$$\therefore H, aKa^{-1} \leq G$$

By Lagrange's Theorem,

$$|HaKa^{-1}| = \frac{|H| |aKa^{-1}|}{|H \cap aKa^{-1}|}$$

$$\Rightarrow |HaK| = \frac{|H| |K|}{|H \cap aKa^{-1}|} \quad (\because |aKa^{-1}| = |K|) \\ \forall a \in G.$$

Q.4: Suppose $H \leq G$ such that whenever $Ha \neq Hb$ then $aH \neq bH$. Show that H is normal in G .

$$\text{Proof: } f: HaK \longrightarrow HaKa^{-1} \\ h_1 a k_1 \longmapsto h_1 a k_1 a^{-1}$$

$$\text{Let } f(h_1 a k_1) = f(h_2 a k_2) \Rightarrow h_1 a k_1 a^{-1} = h_2 a k_2 a^{-1}$$

$\therefore f$ is injective

By fact G is a finite group.

f is onto

Thus, f is bijective:

Let $H \leq G$.

Take any $h \in H$ and $g \in G$.

To show, $H \trianglelefteq G \Rightarrow ghg^{-1} \in H$.

Observe that, $gH = ghH$. ($\because h \in H$)

Hence, $aH = bH$ holds for $a = g, b = gh$. ($\because ghH = gH$)

By contrapositive form,

$$Hg = H(gh) \quad (\because g^{-1}(gh) \in H)$$

$\therefore Hg = Hgh$. Right multiplying by g^{-1} , we get
 $= H = H(ghg^{-1})$

$$\Rightarrow ghg^{-1} \in H. (ghg^{-1}) = (Hgh)g^{-1} = Hgg^{-1} = H.$$

$$\therefore ghg^{-1} \in H.$$

$$\therefore H \trianglelefteq G.$$

Q.5: Let G be a finite abelian group such that the number of solutions of $x^n = e$ is at most n for every positive integer n . □

Show that G is cyclic.

Sol: Let $n \in \mathbb{Z}^+$.
 we know,

$\phi(n) :=$ no. of positive integers $< n$ and relatively prime to n .

Theorem: If G is a cyclic group and $d | n$, where $|G| = n$. Then, no. of elements of order d in G is $\phi(d)$.

~~Assignment-4~~

Theorem: If $m \in \mathbb{Z}^+$ then,

$$m = \sum_{d|m} \phi(d)$$

Example: $m = 8 = 2^3$; $\phi(m) = 2^3 - 2^2 = 4$

$$\therefore \phi(1) + \phi(2) + \phi(4) + \phi(8) = 1 + 1 + 2 + 4 = 8 = m.$$

$$\therefore m = \sum_{d|m} \phi(d)$$

Proof of Q.5: Suppose G has an element such that $O(a) = d$.

Claim: No. of elements of order $d = \phi(d)$

Consider, $\langle a \rangle = \{1, a, a^2, \dots, a^{d-1}\}$ which satisfies, $x^d = 1$.

Assume $|G| = n$.

Suppose, G has no element of order n .

$\therefore$ Let G has elements of order $a_1, a_2, \dots, a_t$. ($a_i \neq n$)

$$\therefore |G| = \sum_{\substack{d|n \\ d \neq n}} \phi(d) = \phi(a_1) + \phi(a_2) + \dots + \phi(a_t) = n$$

$$\text{But } |G| = \phi(a_1) + \phi(a_2) + \dots + \phi(a_t) + \phi(n) + k \text{ for some } k \text{ at most } n.$$

Which is a contradiction.

Thus, G has an element of order $n = |G|$.
 $\therefore G$ is cyclic.

Assignment - 4

Q.1: Show that $\mathbb{Z}_4 \oplus \mathbb{Z}_6 \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{12}$

Ans: We know that the cyclic groups are decomposed into direct sums of cyclic groups of prime power order.

$$\mathbb{Z}_6 \cong \mathbb{Z}_2 \oplus \mathbb{Z}_3$$

$$\text{LHS} = \mathbb{Z}_4 \oplus \mathbb{Z}_6 = \mathbb{Z}_4 \oplus (\mathbb{Z}_2 \oplus \mathbb{Z}_3)$$

$$\Rightarrow \mathbb{Z}_2 \oplus (\mathbb{Z}_4 \oplus \mathbb{Z}_3) \quad (\text{Rearranging by commutativity of direct sum})$$

By using Chinese remainder theorem, $\gcd(4, 3) = 1$, (The direct sum of cyclic groups of coprime orders is cyclic of the product order)

$$\mathbb{Z}_4 \oplus \mathbb{Z}_3 \cong \mathbb{Z}_{12}$$

$$\therefore \Rightarrow \mathbb{Z}_2 \oplus \mathbb{Z}_{12}$$

$$\text{Clearly } \mathbb{Z}_4 \oplus \mathbb{Z}_6 \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{12}$$

Q.2: Show that if G is a finite group and H is a proper subgroup of G , then G cannot be written as a union of conjugates of H .
Burnside's Lemma:

$$\left[\text{no. of orbits} = \frac{1}{|G|} \sum_{g \in G} |F(g)| \right]$$

where $F(g) = \{x \in S : g \cdot x = x\}$ = set of elements of S fixed by $g \in G$

G : finite, $H \leq G$ (subgroup)

$$|H| \mid |G| : |H| = |G|$$

$$|G : N_G(H)| \leq |G : H|$$

Each conjugacy class has

$$|H| \mid |G : N_G(H)| \leq |G|$$

$$|G / N_G(H)|$$

$|H|$ elements
i.e. $\# H \neq \text{conj. classes}$
 $\Rightarrow$ each conj. class

$G = \cup \text{conj. classes}$ only if conj. classes are disjoint

Let S is a G -set
i.e. $S \cong G$

$$1) 1 \cdot a = a \quad \forall a \in S$$

$$2) (g_1 g_2) a = g_1 (g_2 a) \quad \forall g_1, g_2 \in G, a \in S$$

$$a, b \in S, a \sim b \Leftrightarrow a = gb \text{ for some } g \in G$$

$\sim$ (equivalence relation)

$[a]$ = orbit of a .

$$= \{b \in S \mid b \sim a\}$$

$$= \{b \in S \mid b = ga, g \in G, a \in S\}$$

$$= \{ga \mid g \in G, a \in S\}$$

If S is a G -set and $g \in G$

$$\therefore F(g) = \{a \in S \mid ga = a\}$$

Proof: Assume, $G = \bigcup_{g \in G} gHg^{-1} = \bigcup H = H$
(~~$\bigcup gHg^{-1}$~~)

$\Rightarrow H$ is not normal in G .

Because it contradicts $H \leq G$.

So, our supposition was wrong.

$$G \neq \bigcup_{g \in G} gHg^{-1}$$

So, H is not normal in G .

Define $S = \{gHg^{-1} : g \in G\}$

Let S is a set,

$$S \cong G$$

such that

$$g \cdot (x H x^{-1}) = (gx) H (gx)^{-1} \\ = gx H x^{-1} g^{-1} \in S$$

$$[x H x^{-1}] = \{ y(x H x^{-1}) \mid y \in G, x H x^{-1} \in S \} \\ = \{ yx H x^{-1} y^{-1} \mid y \in G \} \\ = \{ t H t^{-1} \mid t = yx \in G \}$$

$\therefore$ There is only 1-~~free~~ orbit.

$$|F(1)| = |S| \geq 2 \quad \left(\because \text{it can't be one (1) because it makes } H \text{ normal as } S = g H g^{-1} = H \neq \right)$$

Claim: $\exists g_0 \in G$ such that $g_0 \neq e$ and $|F(g_0)| = 0$

Proof: Suppose, $|F(g)| \geq 1 \quad \forall g \in G$.

By Burnside,

$$1 = \frac{1}{|G|} \sum_{g \in G} |F(g)| \geq \frac{1}{|G|} [2 + \underbrace{1 + 1 + \dots + 1}_{|G|-1}] \\ \geq \frac{1}{|G|} [2 + |G| - 1] \\ \geq 1 + \frac{1}{|G|}$$

which is not possible.

Suppose, $g_0 \in G$ such that $g_0 = y H y^{-1}$ for some $h \in H$ & $y \in G$

$$\text{Now, } g_0 \cdot y H y^{-1} = g_0 y H (g_0 y)^{-1} \\ = g_0 y H y^{-1} g_0^{-1}$$

$$= y H y^{-1} y H y^{-1} y H y^{-1} \\ = y H y^{-1}$$

Contradiction, as $|F(g_0)| = 0$

Q-5: Show that if $G/Z(G)$ is cyclic, ~~then~~ then G is abelian.

Sol: Suppose, $H = Z(G) = \{ g \in G \mid ga = ag \quad \forall a \in G \}$

$\therefore G/H$ is cyclic.

$$\therefore G/H = \langle gH \rangle$$

Let $x, y \in G$ to show that $xy = yx$.

$$\therefore xH, yH \in G/H = \langle gH \rangle$$

then, $xH = g^r H$; $yH = g^s H$ for some $r, s \in \mathbb{Z}^+$

$$\Rightarrow x = g^r p; \quad y = g^s q \quad (\text{where } p, q \in H)$$

$$\text{Then, } xy = (g^r p) \cdot (g^s q)$$

$$= g^r g^s \cdot p q \quad (\text{as } p \in Z(G) \therefore pg = gp \quad \forall g \in G)$$

$$= g^{r+s} \cdot p q$$

$$\text{Similarly } y \cdot x = (g^s q) (g^r p)$$

$$= g^s g^r \cdot q \cdot p$$

$$= g^{s+r} \cdot (pq) \quad (\because q \in Z(G))$$

$\therefore xy = yx \quad \therefore G$ is abelian.

Assignment-5

Q.3: Show that any finite group with more than 2 elements has a non-trivial automorphism.

Sol: Case I: If G is abelian and $\exists$ some x such that $x \neq x^{-1}$

$f: G \rightarrow G$
 $f(x) = x^{-1}$ } $\rightarrow$ show that it's bijective & homomorphism.

Case II: Assume that $x = x^{-1} \forall x \in G$.

$$G = (\mathbb{Z}_2)^n = \mathbb{Z}_2 \oplus \dots \oplus \mathbb{Z}_2$$

Theorem: If G is a finite abelian group then G is direct sum of cyclic group

$$G = \mathbb{Z}_{p_1^{\alpha_1}} \oplus \mathbb{Z}_{p_2^{\alpha_2}} \oplus \dots \oplus \mathbb{Z}_{p_k^{\alpha_k}}$$

for some prime, $p_1, p_2, \dots, p_k$ (may not be distinct)
 (some $\alpha_1, \dots, \alpha_k$ may not be distinct)

$$G \xrightarrow{f} G$$

$$(x_1, x_2, \dots, x_n) \rightarrow (x_2, x_1, \dots, x_n)$$

it is one-one, onto, homomorphism.

Case II:

```

    G
   / \
abelian non-abelian
  
```

Suppose G is non-commutative.

$\Rightarrow \exists$ some $a, b \in G$ such that $ab \neq ba$

Define $T: G \rightarrow G$ such that

$$T(xy) = a xy a^{-1}$$

$$= \underline{a x a^{-1}} \underline{a y a^{-1}}$$

$$= T(x) T(y)$$

$T(x) = a x a^{-1}$
 (check injective, surjective, homomorphism)

$$\text{Ker } T = \{ x \in G \mid a x a^{-1} = 1 \}$$

$$= \{ x \in G \mid a x = a \}$$

$$= \{ x \in G \mid x = 1 \} = \{ 1 \}$$

$\Rightarrow T$ is injective.

$T(b) \neq b$ Because if $T(b) = b$ then,
 $a b a^{-1} = b$

$\rightarrow ab = ba$ $\times$
 ($\therefore$ contradiction)

Assignment-5

Q.3: Show that $\mathbb{Z}$ is not isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$.

Sol: Suppose $f: \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}$ is an isomorphism.

Then,
 $\exists$ some $a, b \in \mathbb{Z}$ such that $f(a) = (1, 0)$, $a \neq 0$
 & $f(b) = (0, 1)$, $b \neq 0$

$$\therefore f(ab) = f(\underbrace{b + \dots + b}_{a \text{ times}}) = \underbrace{f(b) + \dots + f(b)}_{a \text{ times}} = a f(b) = (0, a)$$

and, also

$$f(ab) = f(\underbrace{a + \dots + a}_{b \text{ times}}) = \underbrace{f(a) + \dots + f(a)}_{b \text{ times}} = b f(a) = (b, 0).$$

Clearly, it contradicts as $(0, a) \neq (b, 0)$.

$\therefore \mathbb{Z}$ is not isomorphic.

Q.4: Let G be a group of order p^n and $p > n$.
Show that if G has a subgroup H of order p then H is normal.

(Theorem: If G is a group and H is only subgroup of $|H|$ in G . Then H is normal.
($\because |gHg^{-1}| = |H|$
 $gHg^{-1} = H$.)

Sol: Let H and K be subgroups of order p .

$$|HK| = \frac{|H||K|}{|H \cap K|} = \frac{|H||K|}{1} \quad (\because |H \cap K| \neq p)$$

$$= \frac{p \cdot p}{1} > p^n \quad (\because \frac{p^2}{1} > p^n)$$

which is not possible

$$\therefore HK \not\subseteq G \quad \text{or} \quad |HK| < |G|.$$

which is contradiction.

Alternate: $|G| = p^n$ $p > n$.

Sylow p -subgroup have order p .

$$n_p = 1 + kp \mid n$$

If $k \geq 1$ then, $1 + kp \geq 1 + p > p > n$. (contradiction)

then, $k=0$, $\therefore \boxed{n_p = 1}$

Q.1: Consider the action of the group of UTM in $GL_n(\mathbb{R})$ acting on $\mathbb{R}^n - \{0\}$. Is this action

1) Faithful

2) Free?

3) Transitive.

Consider an UTM,
$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & a_{nn} \end{pmatrix}$$

$$\therefore AX = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ 0 & \dots & \vdots \\ \vdots & \ddots & \vdots \\ 0 & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

Suppose $G \hookrightarrow S$ then this action is faithful if $gx = x \quad \forall x \in G \Rightarrow g = e$

Suppose, $g \in G$, $T_g: S \rightarrow S$.
 $T_g(x) = gx$.
 T_g is bijective.

$G \not\rightarrow \text{Aut } S$; $g \mapsto T_g$ (check ϕ is homo.)

If ϕ is 1-1 then action is faithful.

$$\begin{aligned} \ker \phi &= \{g \in G \mid T_g = \text{id}_S\} \\ &= \{g \in G \mid T_g(x) = x \quad \forall x \in G\} \\ &= \{g \in G \mid gx = x \quad \forall x \in G\} \\ &= \{g \in G \mid g = e\} \end{aligned}$$

We say, that action is transitive, if there is only 1-orbit

Sol: (iii) $U \rightarrow U$ TM collection.

$$U \hookrightarrow \mathbb{R}^n - \{0\}$$

$$\begin{pmatrix} a_{11} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

e_1 e_2

$a_{11} = 0$ (which is not possible in $GL_n(\mathbb{R})$)
 $\rightarrow$ it'll become non-invertible

Not Transitive.

1. Faithful.

$$\begin{pmatrix} a_{11} & \dots & a_{1n} \\ 0 & \dots & \vdots \\ \vdots & \ddots & \vdots \\ 0 & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

e_1 e_1

$a_{11} = 1$ (if you take $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = e_1$)

and test

=

Construct using each e_j ,
 and set diagonal entries as 1,
 with all UTM upper entries as 0,
 $a_{ij} = 0$ ($j > i$)

Ass. 4

Q.2: $\left(\begin{matrix} \# \text{ elements} \\ \text{in each conj.} \\ \text{classes} \end{matrix} \right) (\# \text{ conjugates}) \leq |G|$

$\Rightarrow G = \bigcup \text{conj. classes}$ only if conj. classes are disjoint. Not true since they intersect in e .